Kire Kolaroski

Senior Machine Learning Engineer

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Open to fully remote roles · 8 years' experience working US hours (UTC+1, overlapping ET/PT)

SUMMARY

Machine learning engineer with 8 years building production ML systems end to end — from ambiguous problem brief to deployed platform. Owned or co-owned three full-lifecycle platforms across time-series forecasting, computational biology, and medical imaging, the last five years fully remote with US clients. Specializes in problem spaces where the modeling approach isn't defined in advance: translating domain problems into systems that biologists, clinicians, and analysts actually use, with expert feedback loops built in. Python, PyTorch, and distributed infrastructure across the full stack — research, modeling, deployment, and delivery.

EXPERIENCE

Machine Learning Engineer / Data Scientist

FatRat · Skopje, North Macedonia (fully remote) · **Jan 2021 – Present**

Lead engineer on ML product engagements for US-based clients, delivering asynchronously across a 6–9 hour time zone gap. Own projects end to end: problem definition, data sourcing, architecture, modeling, deployment, and client delivery.

Biological network analysis and drug property platform — US pharmaceutical client · *Technical lead*

- Architected and built a platform wrapping the **Reactome** pathway database, ingesting its analysis output to identify key intervention nodes required to return a perturbed biological network to equilibrium.
- Designed and implemented the **heuristic graph search algorithms** underlying node identification, producing a ranked set of candidate interventions reviewed and validated by client-side biologists in an iterative feedback loop.
- Grew the engagement from an initial polygenic risk scoring brief into a full analysis platform as client requirements developed.
- Built a parallel pipeline predicting molecular properties from **SMILES** representations covering **ADMET** endpoints, with a focus on metabolism and CYP substrate prediction.

Mitosis detection and expert-in-the-loop labeling platform — US biotech client · *Core contributor*

- Built the complete computer vision pipeline: whole-slide image ingestion, 512×512 tiling, and a **semantic segmentation** model (PyTorch) for detecting mitotic figures.
- Identified label quality as the binding constraint on accuracy and built a labeling platform allowing biologists and clinicians to correct model predictions directly.
- Established an **active learning** loop feeding corrected labels back into training; the dataset reached **200,000 labeled samples**, improving model performance from **70% to 84% F1** across successive iterations.

Machine Learning Engineer

causaLens · London, UK · **Jan 2018 – Dec 2020** *(on-site London, transitioned to remote in 2020)*

First engineering hire at a time-series forecasting startup; the team grew from 5 to 40 people and the company raised **$6M** over the same period. Scope spanned research, core platform engineering, client modeling, and hiring.

Automated time-series forecasting platform (AutoML) — *Core contributor*

- Core contributor to a general-purpose forecasting platform covering data loading, cleaning, automated feature generation, baseline model generation, **hyperparameter optimization**, and automated model summary reporting.
- Implemented **Cython** components for high-throughput streaming data generation feeding the forecasting engine.
- Designed and built the model scoring and **model explainability** subsystems.
- Implemented **HDF5** and CSV ingestion layers for the platform's data pipeline.

Client modeling and delivery

- Worked directly with financial institutions and enterprise clients to translate business problems into forecasting models; delivered solutions across **equities, shipping, and gambling**.
- Designed and implemented **neural network architectures** (LSTM, CNN) for multivariate time-series prediction.
- Ran technical interviews for data scientist and ML engineer hires.

Research Intern

Macedonian Academy of Sciences and Arts, Computer Science Department · **Mar 2017 – Nov 2017**

- Researched information retention in the aggregation of multiplex networks, using entropy-based measures to quantify information loss after aggregation.

SKILLS

Languages — Python, Cython, SQL, JavaScript, Bash

Machine learning — PyTorch, scikit-learn, NumPy, pandas, time-series forecasting, semantic segmentation, computer vision, active learning, hyperparameter optimization, AutoML, model explainability, heuristic and graph algorithms

Information retrieval and search — Information retrieval, search ranking, BM25, embeddings

MLOps and infrastructure — Kubernetes, Docker, AWS, PostgreSQL, Git, CI/CD

Backend and web — Django, FastAPI, React, REST APIs

Domains — Financial time series, computational biology, cheminformatics, drug discovery, medical imaging, digital pathology

EDUCATION

BSc, Computer Science and Engineering — Ss. Cyril and Methodius University, Skopje · 2013 – 2017 · **GPA 9.1/10**

Mathematics-focused track: calculus, linear algebra, probability, statistics, natural language processing, pattern recognition.